Beastlife AI Customer Care Automation System

This solution has been implemented using two approaches:
(1) a LangGraph-based multi-agent architecture for structured AI orchestration

GitHub Repository

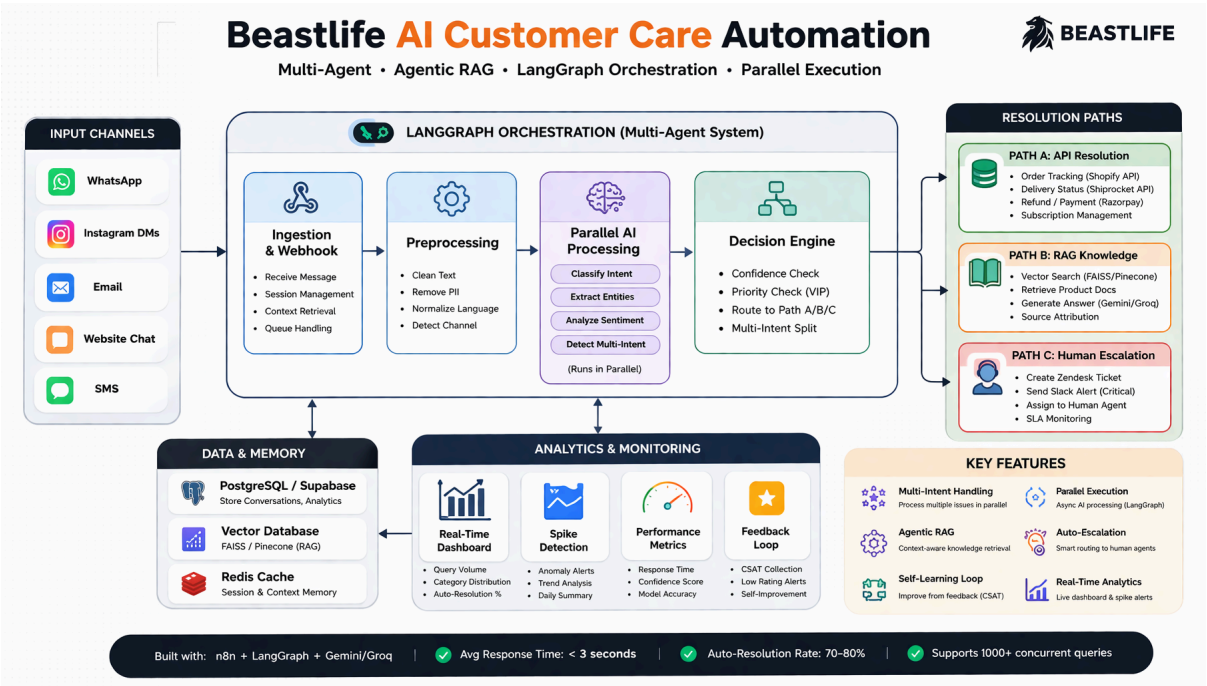
https://github.com/992manav/BeastLife-AI-Customer-Care-Automation-System

Readme+Architecture Repository

https://github.com/992manav/BeastLife-AI-Customer-Care-Automation-System/blob/main/README.md

Dataset Repository

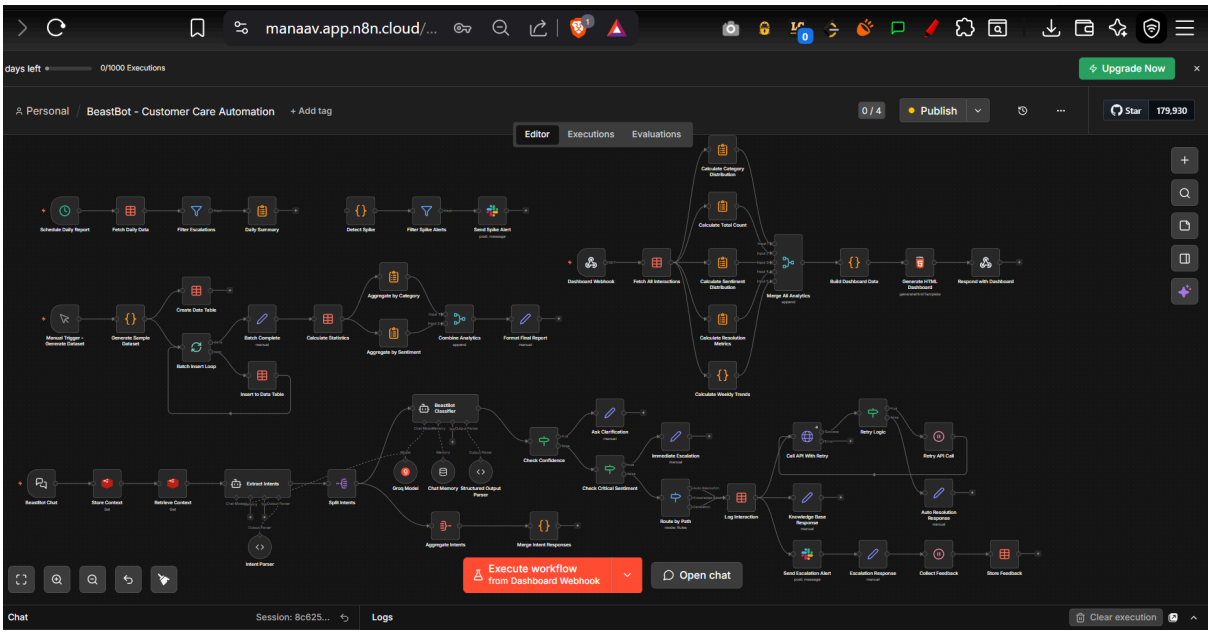
https://github.com/992manav/BeastLife-AI-Customer-Care-Automation-System/tree/main/dataset



(2) an n8n-based workflow automation system for practical, scalable execution.

N8n json

https://github.com/992manav/BeastLife-AI-Customer-Care-Automation-System/blob/main/BeastBot-n8n.json



Requirement Coverage & Enhancements

What Was Required (Assignment Completion)

1. Customer Query Categorization

Developed an AI system to classify customer queries into:

- Order Tracking
- Delivery Delay
- Refund Request
- Product Complaint
- Subscription Issue
- Payment Failure
- General Questions

The system supports inputs from:

- WhatsApp
- Instagram
- Email and Website Chat

---

2. Problem Distribution Dashboard

Built a dashboard that displays:

- Percentage distribution of queries by category
- Most common customer issues
- Weekly and monthly trends

Example insights:

- Order Tracking: 35%
- Delivery Delay: 22%
- Refund Requests: 18%

---

3. Automation Opportunities

Implemented automation for:

- Order tracking responses
- Refund processing
- Payment issue handling
- FAQ-based responses using AI
- Escalation to human support for complex cases

---

4. Tools and Workflow

Technologies used:

- n8n for workflow automation
- Gemini or Groq for AI processing
- LangGraph for orchestration
- FAISS or vector database for knowledge retrieval
- PostgreSQL for data storage
- Streamlit for dashboard visualization

The workflow includes classification, routing, resolution, and analytics logging.

---

5. Deliverables

Provided:

- Workflow architecture (n8n pipeline)
- Sample dataset
- AI categorization logic
- Dashboard for insights
- Scalability explanation

Additional Work Beyond Requirements

Parallel AI Execution

Classification, sentiment analysis, and entity extraction are executed concurrently to reduce response time.

Multi-Intent Handling

The system detects and processes multiple issues within a single query, improving accuracy and resolution.

Agentic Architecture

Implemented a multi-agent system using LangGraph for better decision-making and modular design.

RAG-Based Knowledge System

Used a vector database to retrieve accurate information for product-related queries, avoiding hallucinated responses.

API Integration Design

Designed integration flows for:

- Shopify (order tracking)
- Payment systems

- Logistics providers

Advanced Analytics

Added insights such as:

- Sentiment distribution
- Auto-resolution rate
- Average response time
- Trend analysis

Intelligent Escalation

Implemented priority-based escalation with contextual ticket creation for human agents.

Feedback and Learning Loop

Collected user feedback (CSAT) to continuously improve system performance.

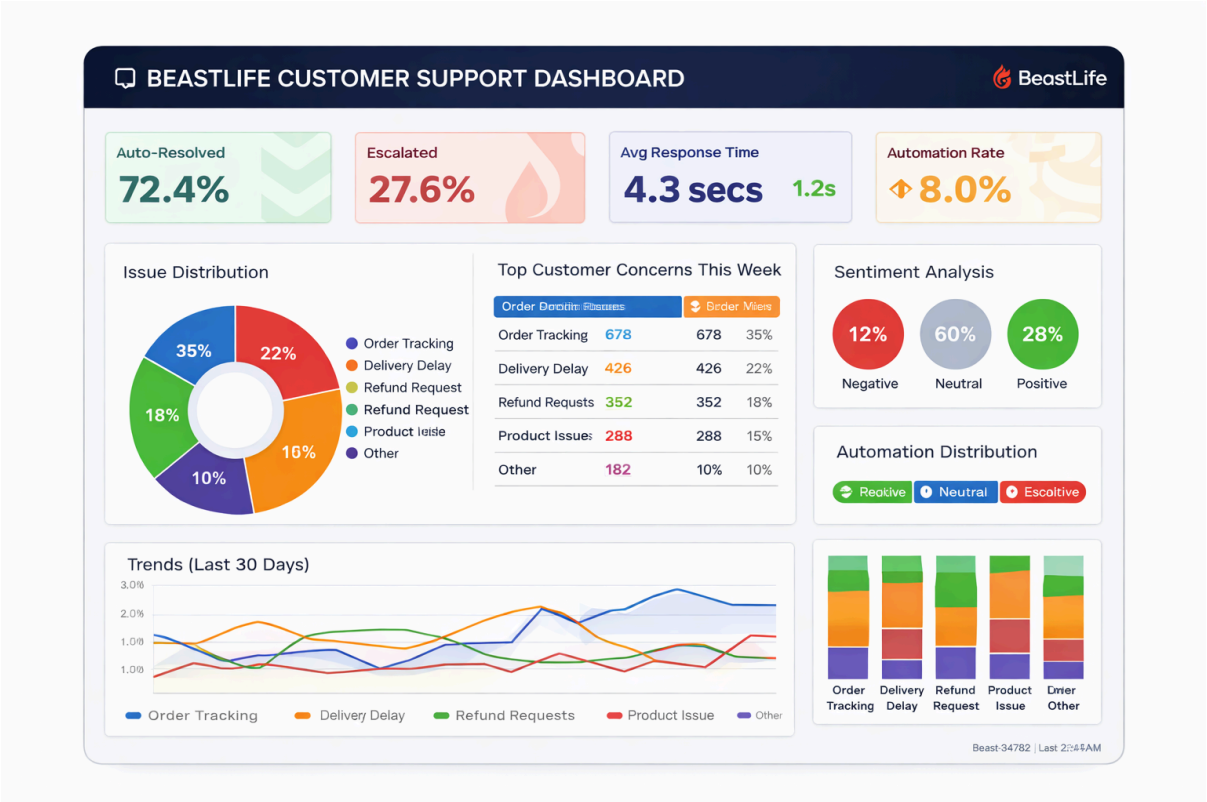
Scalability Design

Designed the system to handle high query volume using asynchronous processing and worker-based execution.

Summary

The system fulfills all assignment requirements and extends them with advanced capabilities such as parallel processing, multi-agent orchestration, and intelligent automation, making it scalable and production-ready.

Dashboard mockup image not made- just created using Figma+Ai



BEASTLIFE AI CUSTOMER CARE SYSTEM

LANGGRAPH + N8N ORCHESTRATION | GEMINI/GROQ | PARALLEL EXECUTION

CHANNELS

WhatsApp | Instagram DM | Email | Website Chat | SMS

LAYER 1 — INGESTION & PREPROCESSING (< 200ms)

Webhook Receiver (n8n Webhook)
Rate Limit • Auth • Logging

→

PII Sanitizer
CARD AADHAAR PAN
→ [REDACTED]

→

Text Normalizer
• Lowercase • Clean
• Hindi/Eng Detect

→

Context Retriever
Redis / PostgreSQL
(Last 5 Messages)

LAYER 2 — PARALLEL AI PROCESSING (Async • 3.2s) — RUNS IN PARALLEL

1. CLASSIFIER — Groq (Llama3-70B)
- Category: order_tracking
- Confidence: 0.92
- All Intents: [tracking, delay]
- Sentiment: negative
- Urgency: high

2. ENTITY EXTRACTOR — Gemini 1.5 Flash
- Order ID: BL-29182
- Product: Whey Protein
- Amount: ₹2,499
- Date: "5 days ago"

3. SENTIMENT ANALYZER — Groq (Mixtral)
- Sentiment: NEGATIVE
- Emotion: Frustrated
- Critical: false
- VIP Check: false

MERGE RESULTS (Wait for All to Complete ~3.2s)

LAYER 3 — DECISION ENGINE (LangGraph Router)

ROUTING LOGIC:

IF confidence < 0.6 → ASK_CLARIFICATION
ELIF sentiment = critical → PATH_C (Escalate)
ELIF category = general_question → PATH_B (RAG)
ELSE → PATH_A (API Resolution)

PATH A — API RESOLUTION (< 2s)

Parallel API Calls

Shopify (Order API) + Shiprocket (Tracking API) + Razorpay (Payment API)

Responses:
- Order: "In Transit - Tomorrow"
- Refund: "₹2,499 Initiated"
- Payment: "Link Generated"
- Subscription: "Cancelled"

PATH B — RAG KNOWLEDGE (~1.5s)

User Query
"Is whey lactose-free?"

↓

Create Embedding
sentence-transformers

↓

Vector Search
Pinecone FAISS (Top 5)

↓

Groq (Llama3) + Context
Answer from docs only • No hallucination

↓

"Whey Concentrate contains lactose. Try Whey Isolate — it has <1g lactose and is suitable for sensitive users."

PATH C — ESCALATION (< 1s)

"Hi Rahul, I've escalated this to our team. They'll contact you within 4 hours. Ref: #BEA-78945"

↓

Create Zendesk Ticket
P1 / P2 Priority

Slack Alert
#support-team

Assign Agent
Specialist Queue

LAYER 5 — ASYNC LOGGING & FEEDBACK (Non-Blocking)

PostgreSQL
- Query Log
- Response Time: 4.7s

→ **Anomaly Detector**
Spike: +250%
(Delivery Issues)

→ **Slack Alert**
#alerts

→ **Schedule CSAT**
(2 hours delay)

→ **Feedback DB**
→ Retrain Weekly
→ Improve Prompts

REAL-TIME DASHBOARD (Streamlit)

By Category	Auto-Resolved	Escalated	Avg Response	Trend (24h)
Tracking 45%	72.4% ↑12%	27.6% ↓12%	4.2s ↓1.3s	1,247 queries